

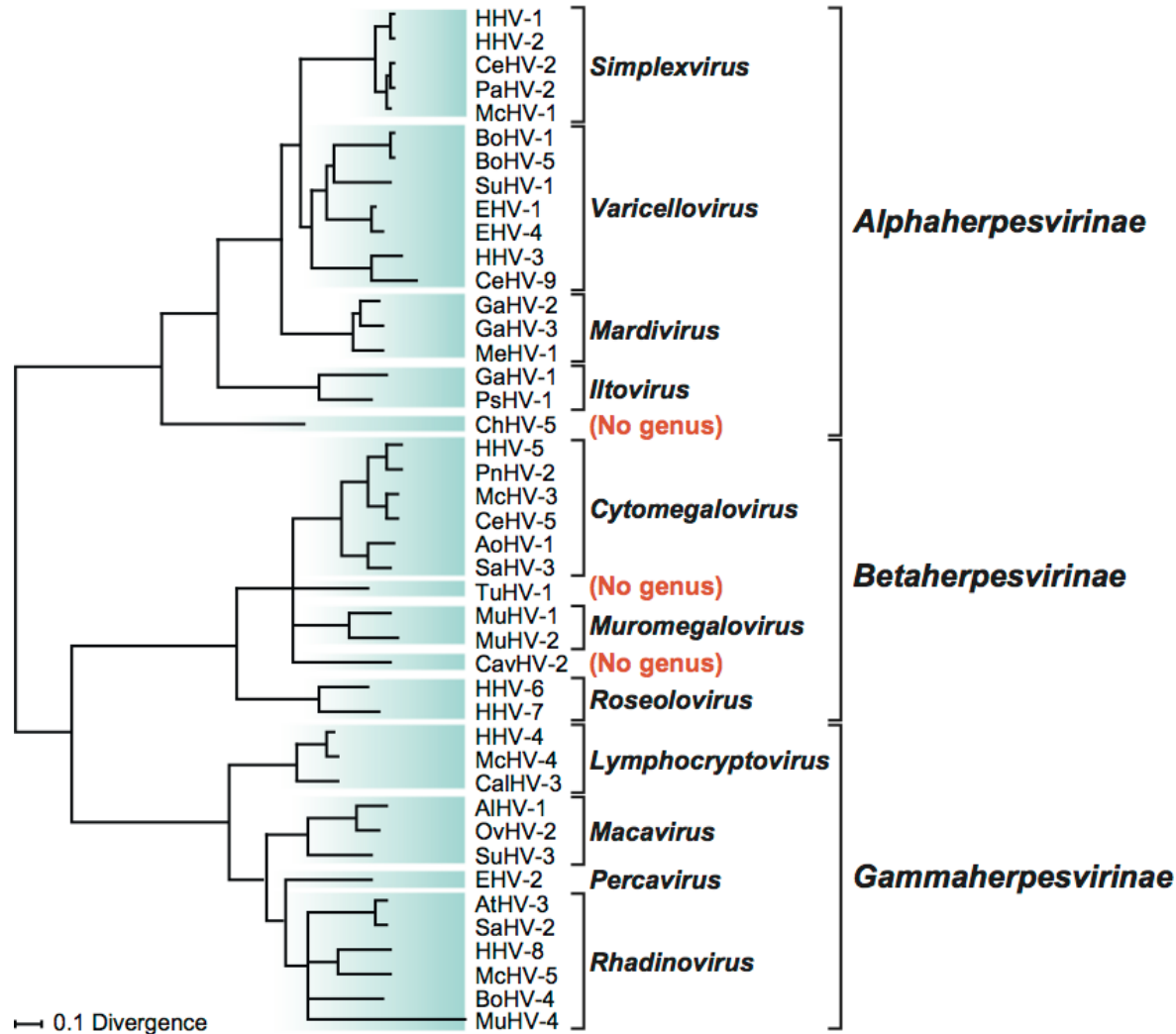
# *FAMILY HERPESVIRIDAE*

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# วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Family *Herpesviridae* ดังนี้
  - ลักษณะและชีววิทยาของไวรัส
  - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
  - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
  - การควบคุมและการป้องกันโรค





# Family *Herpesviridae*

- Current members of family *Herpesviridae* have hosts that include reptiles, birds and mammals
- **Taxonomic structure of the family**
  - Subfamily *Alphaherpesvirinae*
  - Subfamily *Betaherpesvirinae*
  - Subfamily *Gammaherpesvirinae*

# Taxonomic structure of the family

## Subfamily *Alphaherpesvirinae*

- Genus *Iltovirus*
- Genus *Mardivirus*
- Genus *Simplexvirus*
- Genus *Varicellovirus*

## Subfamily *Betaherpesvirinae*

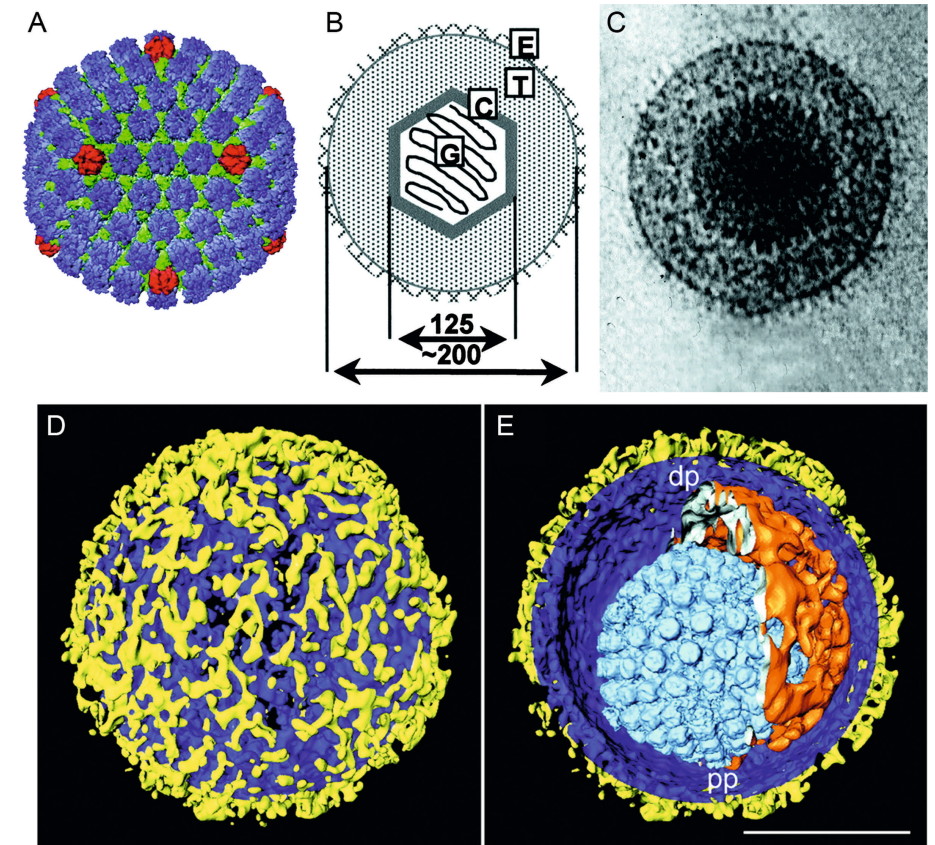
- Genus *Cytomegalovirus*
- Genus *Muromegalovirus*
- Genus *Proboscivirus*
- Genus *Roseolovirus*

## Subfamily *Gammapherpesvirinae*

- Genus *Lymphocryptovirus*
- Genus *Macavirus*
- Genus *Percavirus*
- Genus *Rhadinovirus*

# Virion properties

- The spherical virion is comprised of the core, capsid, tegument and envelope
- Genome: a single molecule, linear, dsDNA (125–295 kbp)
- The mature capsid is an icosahedron
  - 11 pentons and 150 hexons: a total of 161 capsomers (joined by masses, termed triplexes)
- The tegument layers: inner (capsid-associated) and outer (envelope-associated)
  - The tegument contains many proteins, not all of which are required for the formation of virions
- The envelope is a lipid bilayer that is intimately associated with the outer surface of the tegument
  - It contains several different integral viral glycoproteins that form a network of closely spaced spikes
- Enveloped tegument structures lacking capsids can assemble and are released from cells along with virions



Available from: [https://ictv.global/report\\_9th/dsDNA/Herpesvirales](https://ictv.global/report_9th/dsDNA/Herpesvirales)

# Biological and antigenic properties

- The stability of different herpesviruses varies considerably, but they are generally unstable to desiccation and low pH
  - Infectivity is destroyed by lipid solvents and detergents
- Current members of family *Herpesviridae* have hosts that include reptiles, birds and mammals
  - Natural transmission routes: highly contagious aerosol spread, oral contact, sexual contact
  - Most herpesviruses establish a systemic infection, and severe infection is usually observed only in the very young, the fetus, the immunocompromised
  - The key to survival of herpesviruses is their ability to establish life-long latent infection, a feature that is assumed to be the hallmark of all herpesviruses
    - Members of the subfamily *Alphaherpesvirinae* establish latent infection in neurons
    - Members of the subfamily *Betaherpesvirinae* establish latent infection in cells of the monocyte series
    - Members of the subfamily *Gammapherpesvirinae* establish latent infection in lymphocytes
- Infected hosts produce antibodies and cell-mediated immune responses to many structural and non-structural virus proteins
  - Efficient cross-neutralization is observed only between closely related viruses within a genus

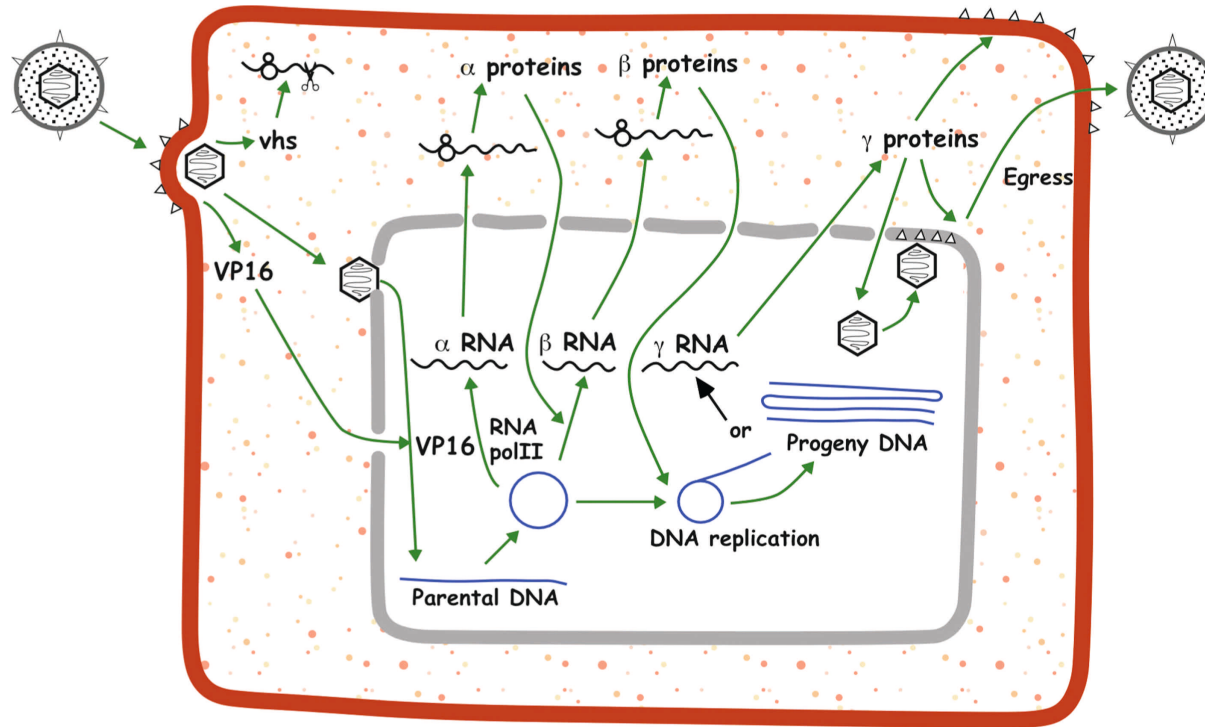
# Virus replication

- **Replicated in nucleus**

- Adsorption and penetration involve the interaction of multiple virion envelope proteins with multiple cell surface receptors
- Entry takes place by membrane fusion either at the cell surface or following endocytosis of the attached virion
- Virion release:
  - Nucleocapsids are observed budding through the inner nuclear membrane into the perinuclear space, become de-enveloped by fusion with the outer nuclear membrane and that the resulting nucleocapsids are re-enveloped in a Golgi or post-Golgi compartment
  - Alternatives include;
    - (i) envelopment at the inner nuclear membrane and subsequent transit to the plasma membrane in transport vesicles
    - (ii) nuclear egress via dilated nuclear pore complexes, followed by envelopment at a cytoplasmic vesicle that is transported to the plasma membrane for virion release

# Virus replication (cont.)

- The nucleocapsid is transported to the region of a nuclear pore, entry of the genome into the nucleus is followed by a transcriptional cascade



- Immediate early ( $\alpha$ ) genes, regulate subsequent gene expression by transcriptional and post-transcriptional mechanisms
  - Immediate early genes can be transcribed in the absence of de novo protein synthesis
- Early ( $\beta$ ) genes encode the DNA replication complex, and a variety of enzymes and proteins involved in modifying host cell metabolism
  - Early gene transcription is dependent on expression of immediate early proteins
- The structural proteins of the virus are encoded primarily by late ( $\gamma$ ) genes
  - Late gene transcription is dependent on viral DNA synthesis

# Herpesviruses infection in animals

**Bovine Herpesvirus 1**

**Canid Herpesvirus 1**

**Equid Herpesvirus**

**Felid Herpesvirus 1**

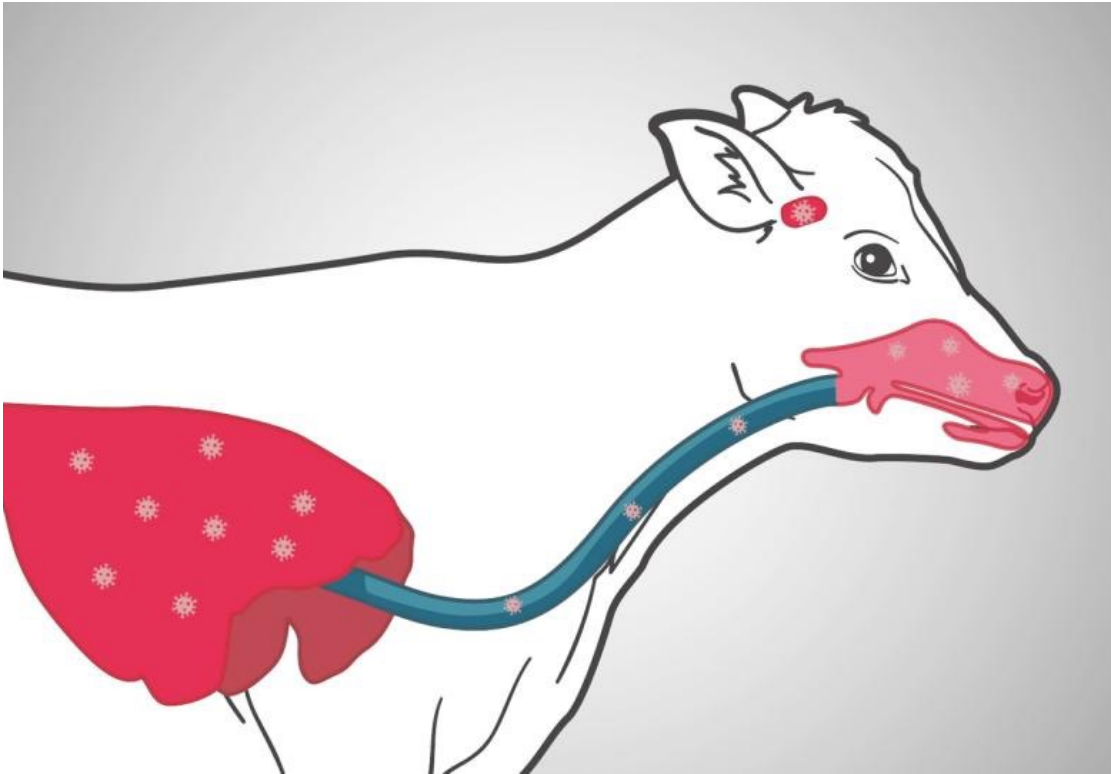
**Gallid Herpesvirus 1**

**Gallid Herpesvirus 2**

**Suid Herpesvirus 1**



# Bovine Herpesvirus 1 (BHV-1)



- **Respiratory (BHV-1.1) and genital (BHV-1.2) diseases**
  - Infectious bovine rhinotracheitis
  - Infectious pustular vulvovaginitis
  - Infectious balanoposthitis
- **Site of latent infection:** sciatic and trigeminal ganglia
- **Pathogenesis and Pathology**
  - Coitus or artificial insemination (genital disease) or droplet transmission (respiratory disease)
  - Incubation period for the respiratory and genital forms of BHV-1 is 2–6 days
  - Viremia is occurred
  - Focal areas of epithelial cell necrosis
  - Tissue inflammation → Pseudo-membrane (accumulation of fibrin and cellular debris)

# Respiratory disease

- Infectious bovine rhinotracheitis: Generalized disease of neonatal calves
  - Fever, depression, inappetence
  - Nasal and ocular discharge (conjunctivitis)
  - Dyspnea, mouth breathing, cough
  - Purulent inflammation (secondary bacterial pneumonia) → hemorrhage → ulcerated mucosa
  - Gastroenteritis
  - Abortion: 4–7 months gestation



# Genital diseases



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- Viral entry by skin and mucus → virus replication; cell lysis occurs → formation of vesicles and ulcers
- Cows: Infectious pustular vulvovaginitis
- Bulls: Infectious balanoposthitis
- **Clinical signs**
  - Fever, depression
  - Tail held away from vulva
  - The vulva is swollen, and small papules, followed by erosions and ulcers, are present on the mucosal surface
  - With bacterial infection, uterine inflammation and transient infertility may occur, with purulent vaginal discharge for several weeks
  - In bulls, similar lesions occur on the penis and prepuce
- If secondary bacterial infections do not occur, animals recover in 10–14 days

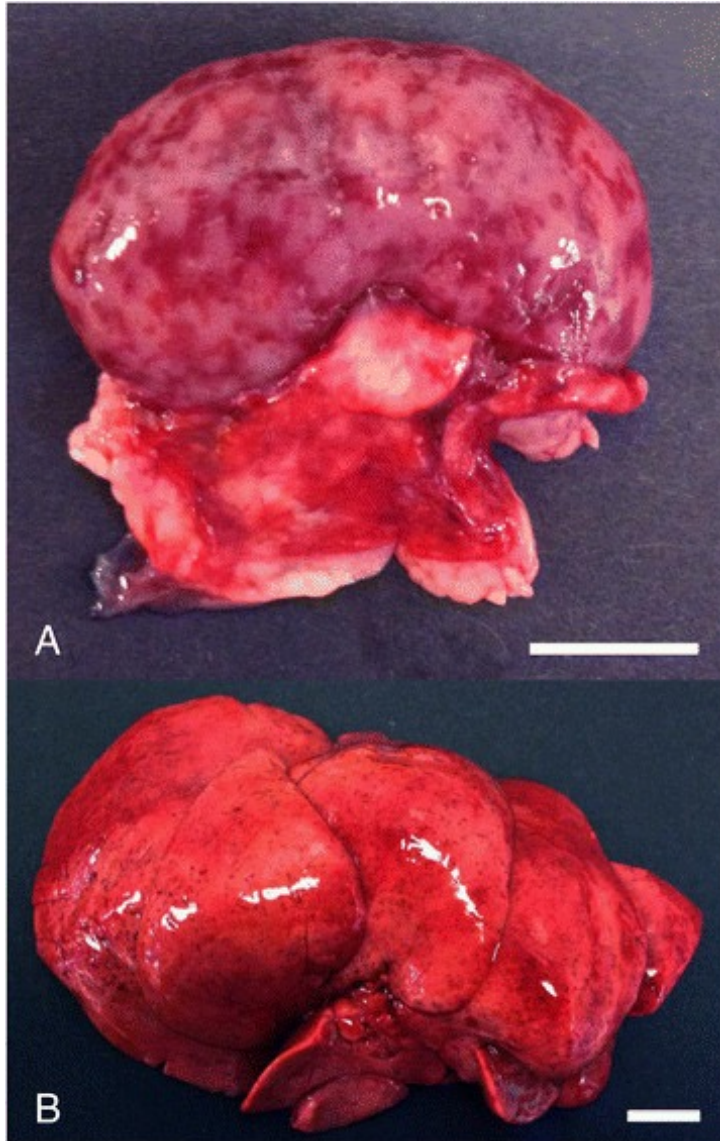
# Diagnosis

- Characteristic lesions: uncomplicated BHV-1 infections can be diagnosed based on characteristic clinical signs and lesions of rhinitis, tracheitis, and conjunctivitis
- Viral isolation; because the severity of disease can vary, BHV-1 should be differentiated from other viral infections by viral isolation
- Paired serological testing; serum antibody titer can be used to confirm a diagnosis
- PCR assay, immunohistochemical testing, or antibody staining
  - PCR assay can be used to identify antigen in a variety of tissues or exudates
  - BHV-1 abortion can be diagnosed by identifying characteristic lesions and demonstrating the virus in fetal tissues by PCR assay, virus isolation, and immunoperoxidase or fluorescent antibody staining



# Treatment and Control

- Treatment: antimicrobial treatment is indicated to prevent or treat secondary bacterial pneumonia in patients with BHV-1
- General recommendations for control;
  - Modified live or inactivated virus vaccines
    - Both parenteral and intranasal modified live virus vaccines are available; however, parenteral vaccines may cause abortion in naive pregnant cattle
    - Intranasal modified live virus vaccines can be used safely in all pregnant cattle
    - Breeding and replacement heifers and bulls should be immunized at 6–8 months old, before breeding, and yearly thereafter
    - Feeder calves should be immunized 2–3 weeks before entry into the feedlot and recommend a second vaccination when animals arrive to provide good duration of immunity
  - Eradication programs; possible by a combination of serological surveillance, culling of reactors, biosecurity, and vaccination



# Canid Herpesvirus 1

- Only canids (dogs, wolves, coyotes) are known to be susceptible
  - The seroprevalence in dog populations worldwide ranges from 20-98%, depending on the region and population studied
- A rapidly progressive and fatal multisystemic disease of neonatal puppies
  - It can result in lifelong latent infections and can lead to mild respiratory and ocular signs in older dogs
- Generalized hemorrhagic disease of pups; under 4 weeks of age
  - Gross necropsy findings; hemorrhage ecchymosis (kidney and gastrointestinal tract)

# Pathogenesis



- CHV is usually transmitted by contact between susceptible individuals and the infected oral, nasal, or vaginal secretions of shedding dogs
  - Immunologically naive pregnant bitches are at risk of acute infection, which may be transmitted to fetuses or neonatal pups
- Canine herpesvirus replicates more efficiently at the lower body temperatures maintained by neonates, deaths due to CHV infection occur usually in puppies 1–3 weeks old
  - After this time, natural resistance to infection improves as puppies mature and maintain a higher body temperature
- Virus replication in the surface cells of the nasal mucosa, pharynx, and tonsils
- In newborn susceptible pups or other immunocompromised dogs, viremia arises, and diverse visceral organs are invaded
- Viral shedding; shedding by latently infected animals after clinical or subclinical, recrudescence is of lesser severity and duration

# Clinical signs

- Clinically, neonatal canine herpesvirus is characterized by the following:
  - Severe nonspecific systemic signs in a neonatal puppy
  - Diarrhea, inappetence, lack of weight gain
  - Tachypnea, pneumonia
  - Corneal edema, conjunctivitis
  - Lack of fever
  - Rapidly progressive illness
  - Failure to thrive or sudden death
- In older dogs; mild upper respiratory tract signs, which may be part of the kennel cough syndrome (infectious tracheobronchitis)

# Diagnosis

- Clinical evaluation
- PCR assays can be used to confirm the diagnosis (used on fresh tissue and fluid samples)
- Necropsy; the focal areas of necrosis and hemorrhage, especially those that occur in the kidneys, distinguish CHV infection from hepatitis and neosporosis
- Virus isolation from fresh lung, liver, kidney, and spleen by cell culture techniques, followed by identification using PCR assay
- Serologic testing
  - Hemagglutination, ELISA, and immunofluorescence antibody tests are available



# Prevention

- A vaccine for Canine Herpesvirus is available in European countries, but not in the United States
- Previously infected bitches generally develop antibodies, and puppies receive maternal antibodies in the colostrum
- Isolation of pregnant bitches from other dogs during the last 3 weeks of gestation and first 3 weeks postpartum, with excellent hygiene by human handlers, is the most effective way to minimize risk to puppies
- CHV is sensitive to lipid solvents (such as ether and chloroform) and to most disinfectants





# Equid Herpesvirus (EHV)

- EHV-1 and EHV-4 are major causes of viral respiratory disease in horses
  - EHV-1 also causes abortion and neurologic disease
- EHV-2 is of questionable importance as a cause of respiratory disease in horses
- Viral latency and a carrier state are associated with EHV infection; this allows the virus to be maintained within populations of horses
- Transmission is primarily via respiratory secretions; diagnosis is based on PCR array of a nasal swab in actively shedding horses and PCR array of whole blood in viremic horses
- Vaccination and appropriate biosecurity practices are essential in controlling outbreaks and preventing the spread of EHV

# EHV infections

- Equid Herpesvirus 1: Equine abortion virus
  - Transmission: respiratory tract
  - Respiratory disease and abortion or encephalitis
    - Aborted fetuses
      - Interstitial pneumonitis
      - Severe necrosis of the spleen
      - Focal necrosis of the liver and adrenal glands
- Equid Herpesvirus 4: Equine rhinopneumonitis virus
  - Acute respiratory disease
    - Fever, anorexia
    - Serous or mucopurulent nasal discharge



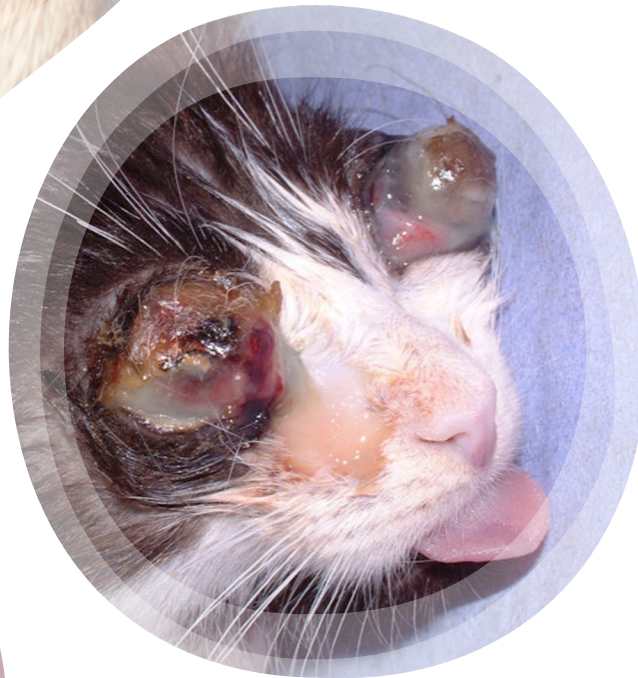
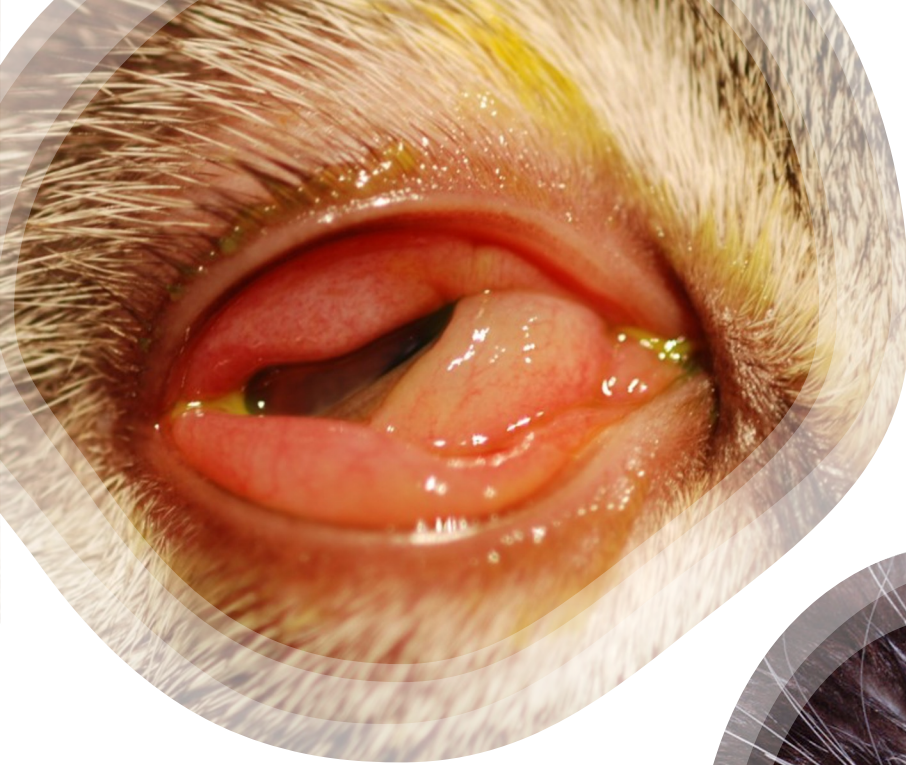
**FIGURE 9.7** Equid herpesvirus abortion: interstitial pneumonia in aborted foal. (Courtesy of H. DeCock, University of California.)

# Felid Herpesvirus 1 (FHV-1)

- FHV-1 causes feline viral rhinotracheitis, an important viral disease of cats on a worldwide basis
- Acute infection is associated with both upper respiratory and ocular signs – following the acute phase of the disease lifelong latency is established, primarily in sensory neuronal cells
  - The trigeminal ganglion is considered a primary site of latency for FHV-1
- Immunization is strategy for protection against FHV-1 and other alpha-herpesvirus infections

# Clinical signs

- Fever, anorexia
- Sneezing and cough
- Serous nasal and ocular discharges
- Frothy salivation
- Dyspnea
- Keratitis: corneal ulcers



# Diagnosis and control

- Clinically, there is an overlap between the symptomatology of acute FHV-1 and feline calicivirus (FCV), another major respiratory disease of cats
- Laboratory diagnosis: tissue homogenates or swabs
  - FA test, virus isolation (gold standard), and PCR
- **Treatment and control**
  - Supportive treatment and antiviral agents (oral administration of famciclovir has been reported to be safe and efficacious in treating ocular signs, cutaneous disease, and rhinosinusitis)
  - Vaccination recommendations have been provided by The European Advisory Board for Cat Diseases (ABCD) and The American Association of Feline Practitioners Feline Vaccine Advisory Panel
    - The ABCD panel recommends an initial two-dose vaccination regimen: the first dose being given at 9 weeks of age and the second at 12 weeks of age, this is followed by yearly boosters
    - The American Association of Feline Practitioners Feline Vaccine Advisory Panel advises that the primary immunization dose should be given as early as 6 weeks of age, with additional doses every 3 to 4 weeks until 16 weeks of age – a booster dose is to be administered 1 year following the last dose of the primary series

# Gallid Herpesvirus 1

## Avian infectious laryngotracheitis

- Most common in 4-18 months of ages

## Clinical signs

- Sneezing and cough
- Nasal and ocular discharge
- Depression
- Dyspnea
- Head shaking
- Pump handle respiration: raised neck and extended head during inspiration
- Fowl diphtheria: asphyxia

## Gross lesions

- Larynx and trachea
  - Hemorrhagic inflammation
  - Edema
  - Ulceration
  - Necrosis
- Formation of diphtheritic membranes



# Gallid Herpesvirus 2

## Marek's disease virus



Lymphoproliferative disease of chickens



Transmission

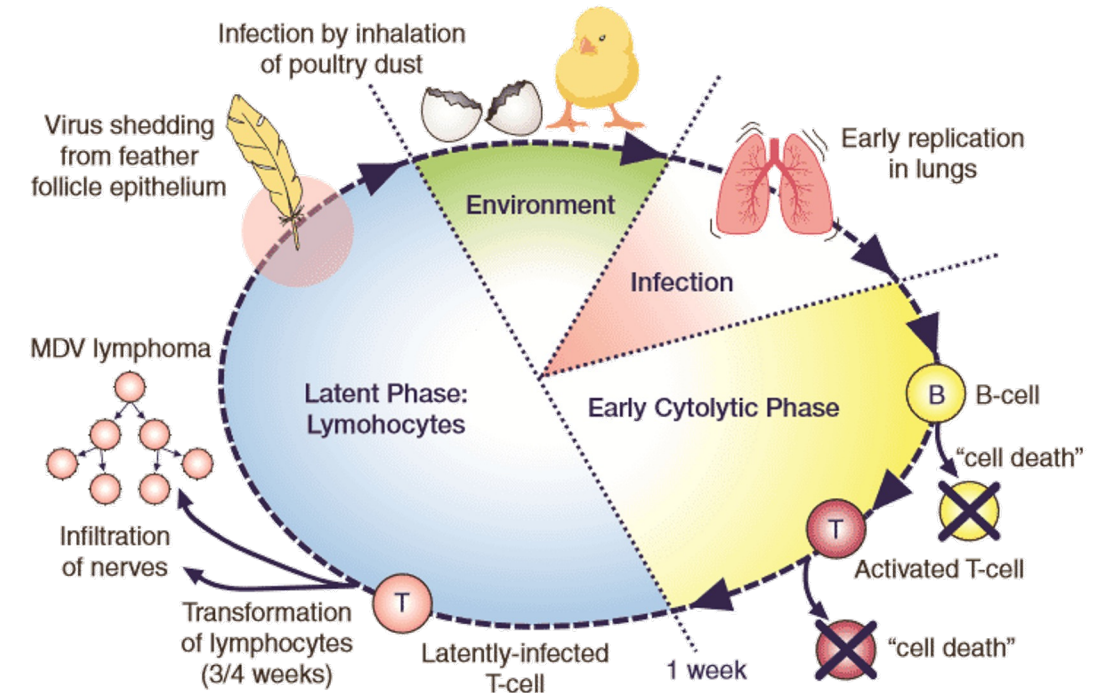
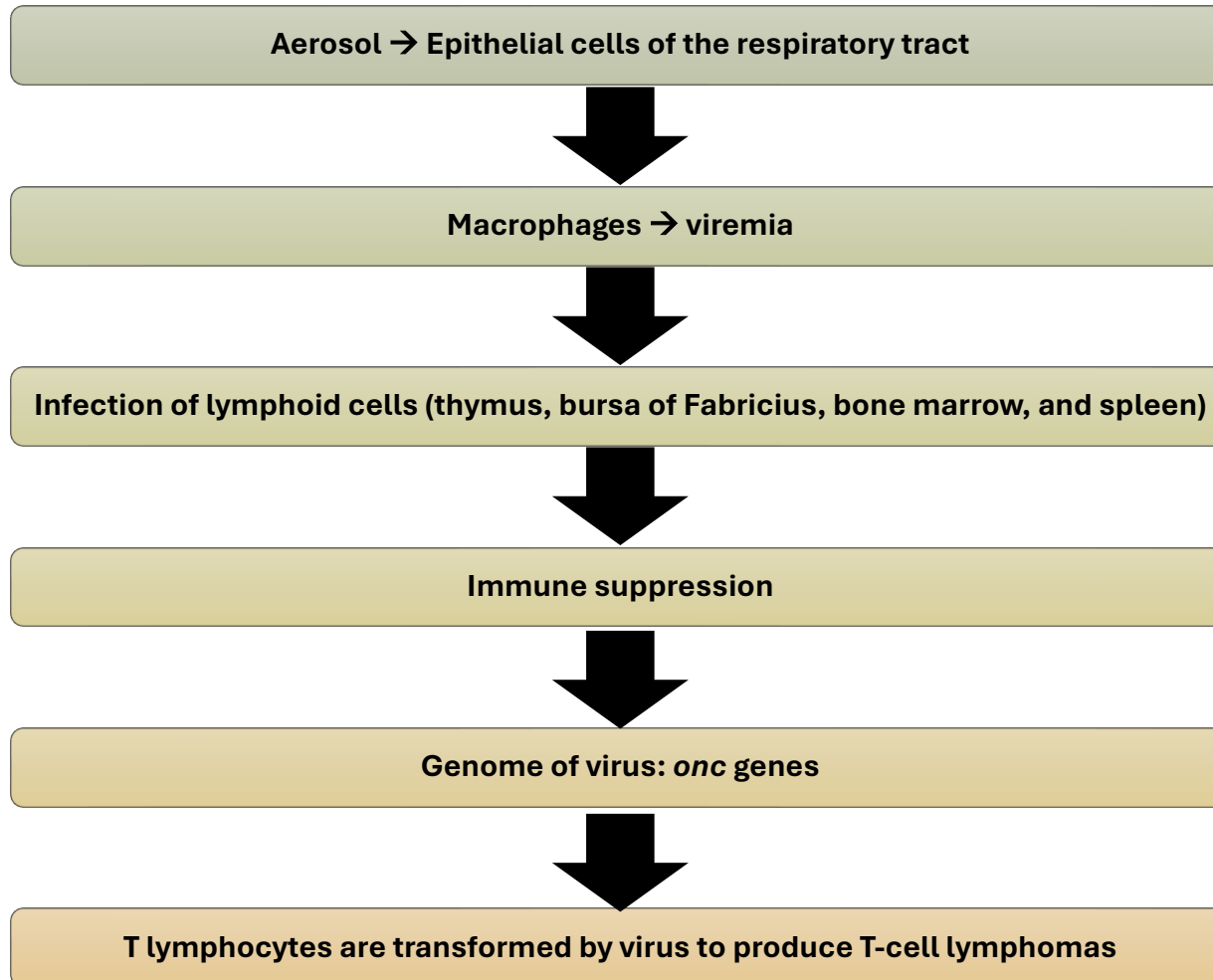
- Direct contact
- Inhalation of dander (epithelial cells)

## Clinical signs

- Classical Marek's disease
  - Lymphomas in visceral organs
  - Asymmetric paralysis of legs or wings
  - Incoordination
- Acute Marek's disease or fowl paralysis
  - Depression
  - Ataxia or paralysis
- Ocular lymphomatosis
- Cutaneous Marek's disease
  - Red leg syndrome



# Pathogenesis and Pathology



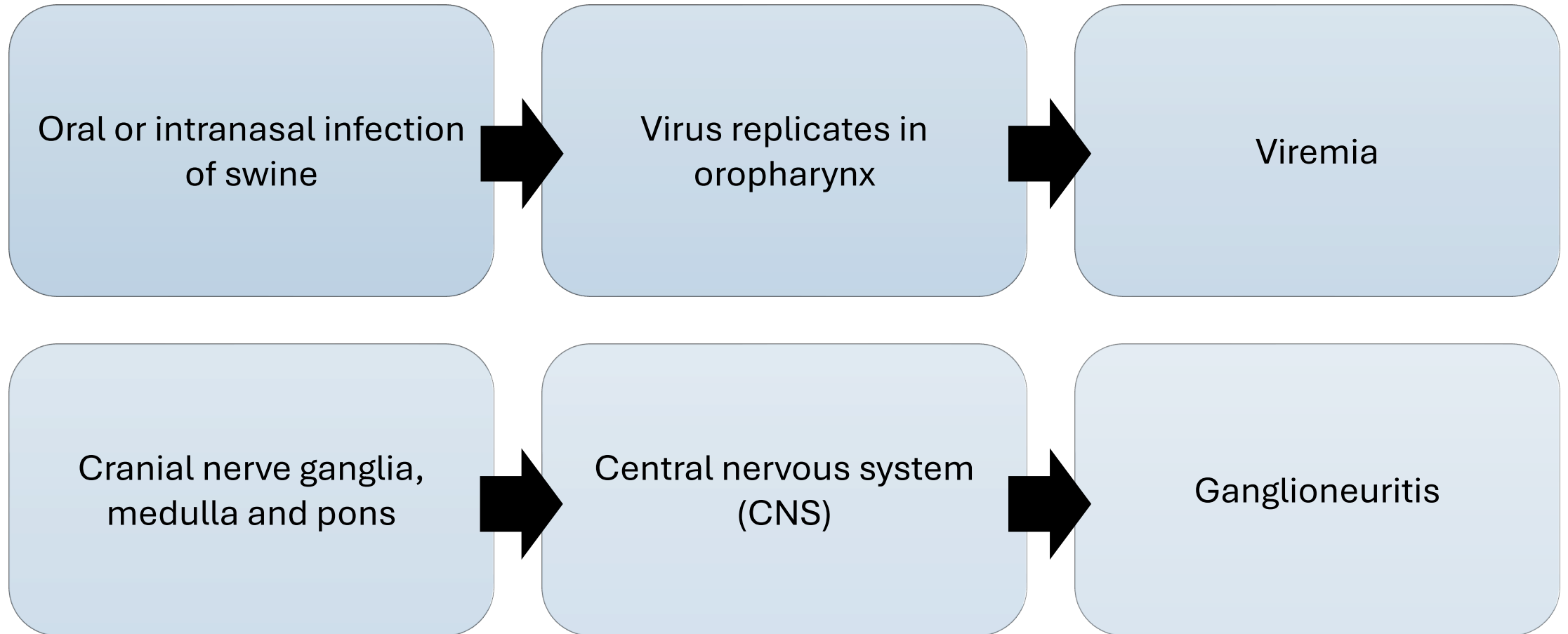
# Diagnosis and control

- **Diagnosis**
  - History, clinical signs and gross necropsy
  - Detection of antigen: FAT or EM
  - Detection of antibody: SN, Gel diffusion or indirect FAT
  - Viral isolation in cell cultures, CAM or yolk sac
- **Control**
  - Vaccination\*\*

# Suid Herpesvirus 1

- **Pseudorabies or Aujeszky's disease virus**
  - Primary host or reservoir host
    - Swine
  - Secondary hosts
    - Horses, cattle, sheep, goats, dogs, cats and many feral species
- **Virus shedding**
  - Saliva and nasal discharges of swine
- **Transmissions**
  - Licking, biting, aerosols, or ingestion

# Pathogenesis and Pathology



# Clinical signs in swine

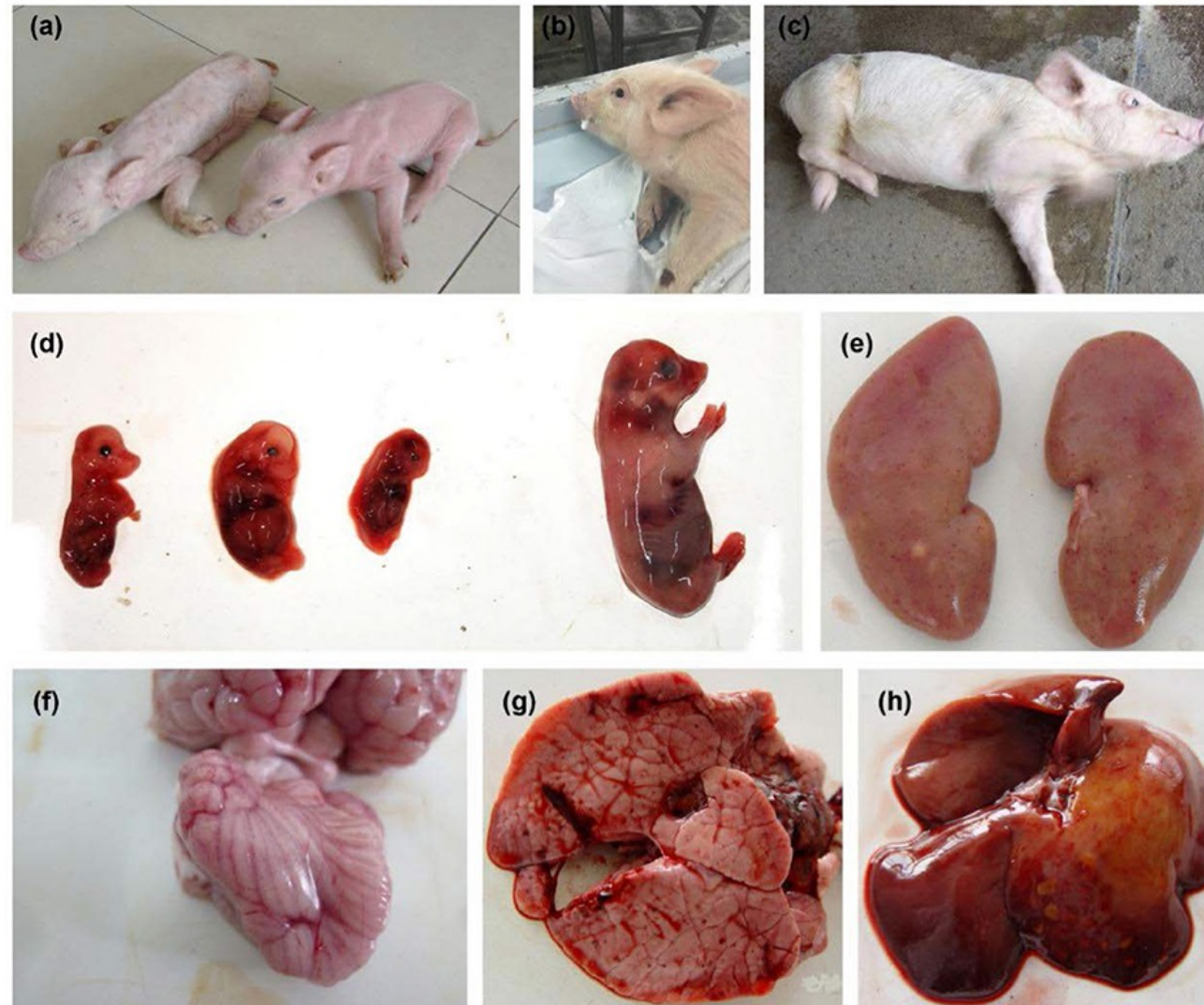
## Pregnant Sows

- Before 30<sup>th</sup> day of gestation
  - Death and resorption of embryo
- After 30<sup>th</sup> day of gestation
  - Abortion
- Late pregnancy
  - Mummified, macerated, weak and stillborn

## Others

- Piglets
  - Maternal antibody
- Weaned, Growing, and Mature Swine
  - Fever
  - Vomiting
  - Incoordination
  - Muscle spasm
  - Circling
  - Convulsions
  - Excess salivation





Zhang C, Cui H, Zhang W, Meng L, Chen L, Wang Z, Zhao K, Chen Z, Qiao S, Liu J, Guo Z, Dong S. Epidemiological Investigation of Porcine Pseudorabies Virus in Hebei Province, China, 2017-2018. *Front Vet Sci.* 2022 Jun 24;9:930871. doi: 10.3389/fvets.2022.930871. PMID: 35812861; PMCID: PMC9263846.

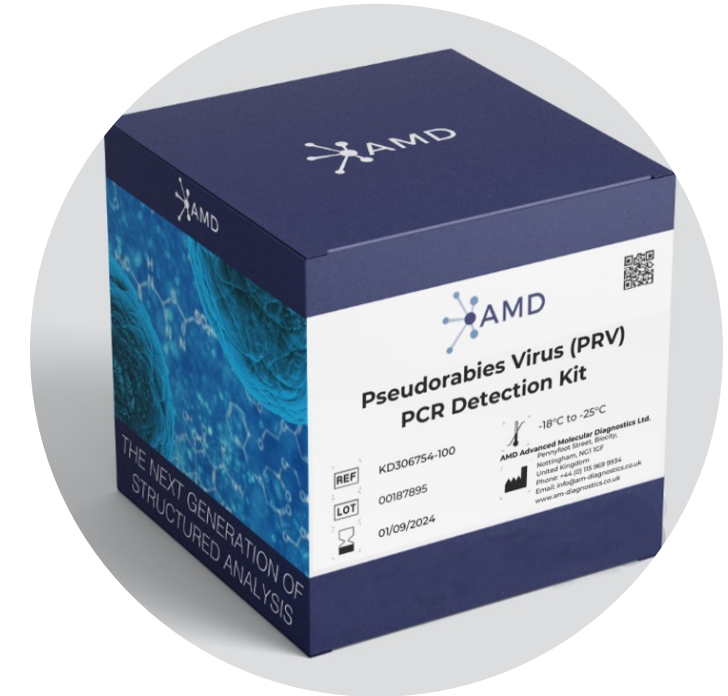


# Clinical signs in secondary hosts

- Cattle: Mad itch
  - Intense pruritus
  - Lick (flanks or hind limbs)
  - Gnawing and rubbing (abrasion)
  - CNS signs
- Dogs and cats: Pseudorabies
  - Intense pruritus
  - Paralysis of the jaws and pharynx
  - Drooling of saliva

# Diagnosis

- History and clinical signs
- Virus isolation
- PCR assay
- Immunohistochemistry and FAT – tissue samples
- Serological tests
  - Serum neutralization
  - ELISA





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- Quinn, P.J., Markey, B.K., Leonard, F.C., FitzPatrick, E.S., Fanning, S. and Hartigan, P.J. 2011. Veterinary Microbiology and Microbial Disease. 2<sup>nd</sup> edition. Wiley-Blackwell, West Sussex, UK. 912p.